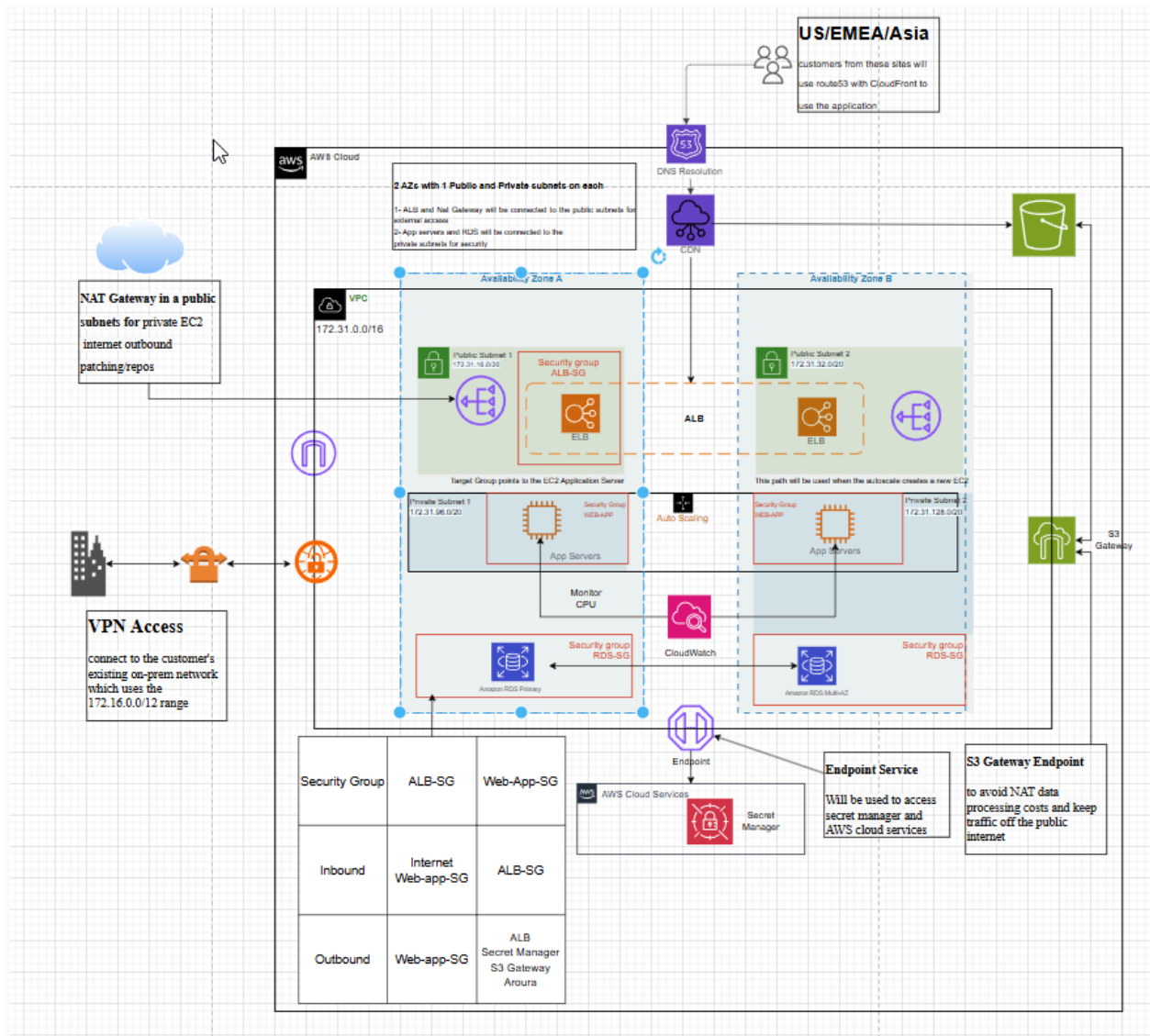


Delivering the required points

1- Diagram/specs should include additional well-architected security considerations (e.g., security groups/NACLs, NAT, VPC endpoints, clear routing boundaries, Secrets Manager placement)



I have added the services below to the diagram:

- security groups
- VPC Endpoint
- S3 Gateway
- Routing rules
- NAT Gateway

- Secret Manager

2- Architecture should reflect decoupling/scaling mechanisms in a well-architected way (e.g., Auto Scaling + CloudWatch, and/or asynchronous options like SQS/SNS/Lambda/EventBridge where appropriate)

- added Cloudwatch service and configured alarms to to send SNS topic to an email for IT team
- Configured the autoscale policy to scale out based at 80% of the CPU load

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template or configuration

Choose launch template or configuration

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.

Name
Auto Scaling group name
Enter a name to identify the group.
food-dis-app-autoscale
Must be unique to this account in the current Region and no more than 255 characters.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
food-dis-app-template
Create a launch template

Version
Default (1)
Create a launch template version

Description
-

AMI ID
ami-00a8151272c45cd8e

Key pair name
-

Launch template
food-dis-app-template
lt-03de2ef5aef6116af

Instance type
t3.micro

Security groups
-

Security group IDs
sg-0f12b3f58325446dd

Request Spot Instances
No

Additional details

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 2: Choose instance launch options

Choose instance launch options

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements
You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.
Launch template: food-dis-app-template
Version: Default
Description: t3.micro

Network
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
vpc-020b269346c19e105
172.31.0.0/16 Default
Create a VPC

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
usw2-az1 (us-west-2b) | subnet-0f9ff73a4eed6e08 (private-subnet-2)
usw2-az2 (us-west-2a) | subnet-03bc1bfb792933b81 (private-subnet-1)
Create a subnet

Adding the 2 private AZs

EC2 > Auto Scaling groups > food-dis-app-autoscale

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager [New](#)

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

Auto Scaling Groups

Settings

Auto Scaling group updated successfully

food-dis-app-autoscale autoscale name

food-dis-app-autoscale Capacity overview [Edit](#)

am:aws:autoscaling:us-west-2:626183958825:autoScalingGroup:18a102d4-c19d-42d7-9c84-57d8096ef58f:autoScalingGroupName/food-dis-app-autoscale

Desired capacity	Scaling limits	Desired capacity type	Status
1	1 - 2	Units (number of instances)	-

Date created
Thu Jan 15 2026 18:58:37 GMT+0200 (Eastern European Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags - moved

Launch template [Edit](#)

Launch template

lt-03de2ef5aef6116af

food-dis-app-template

AMI ID

ami-00a8151272c45cd8e

Instance type

t3.micro

Security groups

-

Security group IDs

sg-0f12b3f58325446dd

App-SG

Owner

arn:aws:sts::626183958825:assumed-role/voclabs/user:2107184-91c6c672-13ee-11ed-a441-0b4c0cb6e8ad

Create time

Thu Jan 15 2026 18:47:30 GMT+0200 (Eastern European Standard Time)

Request Spot Instances

No

Version

Default

Description

-

[View details in the launch template console](#)

AMI ID

ami-00a8151272c45cd8e

Security groups

-

Storage (volumes)

-

Key pair name

-

Network [Edit](#)

Network [Edit](#)

Availability Zones

usw2-az2 (us-west-2a)

usw2-az1 (us-west-2b)

Subnet ID

subnet-03bc1bfb792933b81

subnet-0f9fff73a4eed6e08

Availability Zone distribution

Balanced best effort

Instance type requirements [Edit](#)

Your Auto Scaling group adheres to the launch template for purchase option and instance type.

Health checks [Edit](#)

Health check type

EC2, ELB

Health check grace period

300

Use ELB target groups and EC2 CPU as health checks

Instance maintenance policy [Edit](#)

Replacement behavior

Custom behavior

Min healthy percentage

90

Max healthy percentage

110

Capacity Reservation preference [Edit](#)

Preference

None

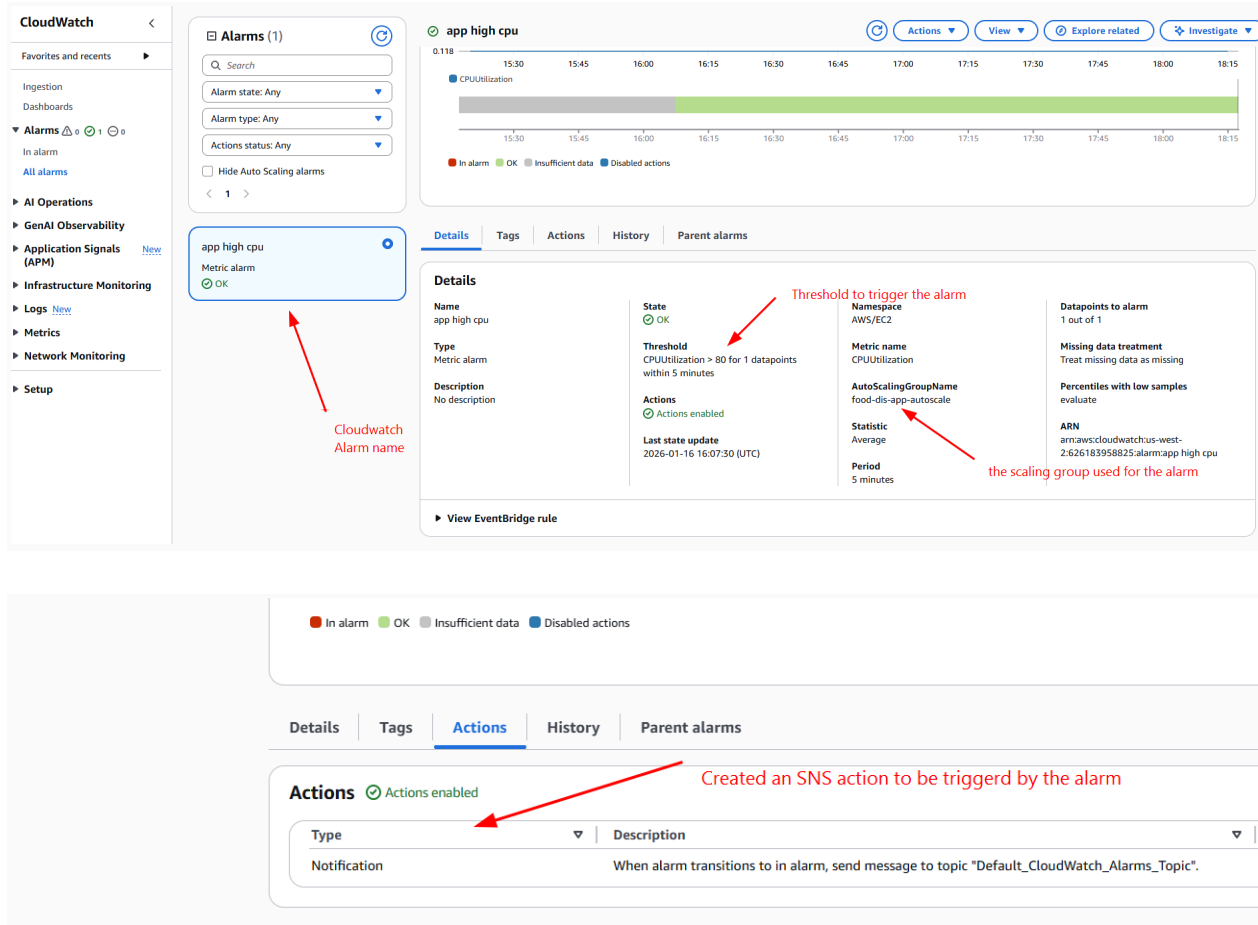
Capacity Reservation IDs

-

Resource Groups

-

Advanced configurations [Edit](#)



3- Pricing/right-sizing recommendations should be consistent, clearly justified, and reflect additional cost-avoidance levers (e.g., endpoints to reduce egress/NAT costs)

I already provided the sizing recommendation on my last submission but for the On-demand option, the recommendation below shows for the Saving Plan option

Assumptions

- Region: us-east-1
- Instance type: r7g.large
- OS: Amazon Linux (Linux pricing)
- Tenancy: Shared
- Pricing model: Saving Plan
- Hours/month: 730
- Instances: 2 (Multi-AZ)
- NAT Gateway: 1 per AZ (730 hrs)
- Data egress: 100 GB/month

I have attached a detailed screenshot showing how i used cost estimator

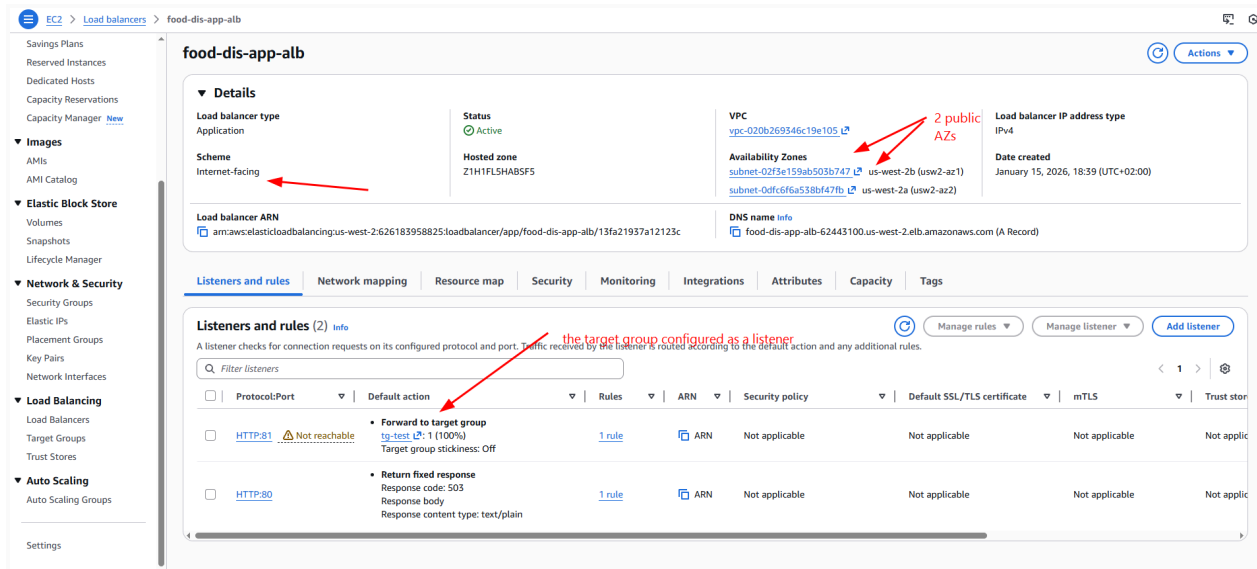
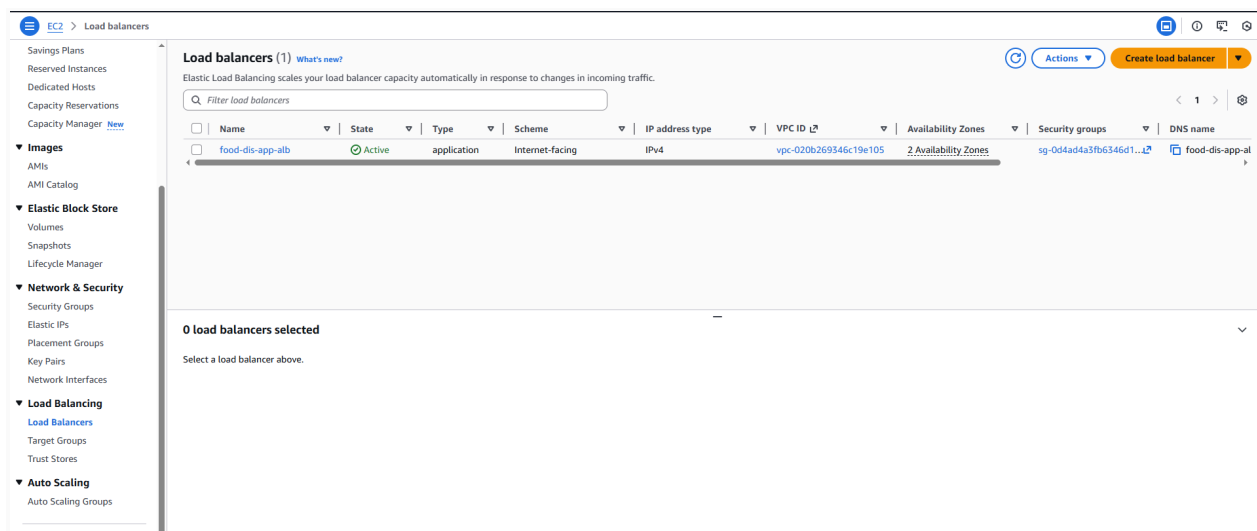
- Monthly cost: 47.58 USD
- Total 12 months cost: 570.96 USD

Justification:

- We will use R7g Instances:
- Processor: AWS Graviton3 (ARM-based)
- Memory: DDR5 with 50% higher bandwidth than previous generations
- High memory-to-vCPU ratio (8:1)
- Excellent for PHP's memory-intensive operations
- Great for nginx caching and buffering
- Cost-effective ARM performance

4- Scaling strategy should include the Availability Triad (ALB + Auto Scaling + CloudWatch) or clearly explain an alternative

- Added Application load balancer which will be in 2 public AZs
- The ALB has target group points to the EC2 instance available on private AZ, the EC2 instance created by the auto scaling group
- The auto scaling group run 1 instance (for cost management, but we can make it 2 VMs)
- The auto scaling group check the ALB healthy targets and CPU metric
- Created a cloudwatch alarm based on the auto scaling group that will trigger an SNS action to the target mail



EC2 > Target groups > tg-test

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

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Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Details

arn:aws:elasticloadbalancing:us-west-2:626183958825:targetgroup/tg-test/651

Target type

Instance

Protocol : Port

HTTP: 80

IP address type

IPv4

Load balancer

food-dis-app-alb

1

Total targets

0

Healthy

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (1)

Target groups route requests to individual registered targets using the protocol and automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

1 match

Health status : unhealthy

Clear filters

Instance ID

Name

Port

i-016e096ead1c74cfa

-

80

due to permission issue

i can't create crt for https

so I chose http